

Austo Motor Data Analysis Business Report

Data-Driven Insights for Austo Automobiles: Customer Preferences and Purchase Patterns

Prepared by: Smriti Shukla

Great learning PGP - DSBA

May 2025

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Introduction

Austo Motor Company is a premier automobile manufacturer renowned for its diverse range of SUVs, Sedans, and Hatchbacks. Recognizing the need to optimize marketing efforts, this report aims to analyse current customer database and sales data to enhance customer targeting and retention. By understanding customer preferences based on demographics, income, and buying behaviour, we can refine our marketing campaigns to reach the most potential customers effectively.

Executive Summary

This comprehensive report delves into company's customer and sales data to identify patterns and insights that can drive strategic marketing decisions. It covers:

- Distribution of vehicle prices, customer ages, and income levels.
- Analysis of vehicle models or makes favoured by different customer segments.
- Sales preferences across various demographics such as age groups, income, professions, and marital status.
- Identification of key customer profiles and market segments.
- Actionable recommendations to tailor marketing strategies for higher engagement and conversion.

Findings will empower stakeholders to make data-driven decisions, ensuring marketing efforts are aligned with customer preferences and maximizing ROI.

Key Highlights

Statistical Summary

- Sales Data: Total vehicles sold, average sales per model/type.
- Customer Age: Distribution across age groups; identification of prime purchasing ages.
- Income & Salary: Range, average, and distribution of customer salaries.
- Loan & Financing: Percentage of customers utilizing loans; impact on purchasing behavior.

Visual Data Representations

- Vehicle Make & Type: Bar charts illustrating sales volume of SUVs, Sedans, Hatchbacks.
- Customer Age Distribution: Histogram highlighting key age groups purchasing different models.
- Customer Preferences: Heat maps correlating age, income, and vehicle preferences.

Customer Segmentation & Preferences

- By Age Group Younger customers tend to prefer Hatchbacks; middle-aged customers favor SUVs.
- By Income Level Higher-income customers predominantly purchase luxury sedans; middle-income groups lean towards Hatchbacks.

By Profession Salaried professionals show distinct preferences compared to self-employed or retired customers.

- Marital Status- Married customers prefer family-oriented SUVs and Sedans.

Key Customer Profiles

- Salaried, middle-aged, married customers with higher income levels: Prefer SUVs and luxury Sedans.
- Young professionals, single, lower-income customers: Favor Hatchbacks and affordable sedans.

Data Overview

Data Source

Customer automobile purchase data collected company sales records in CSV format

Column names

- Gender: Male/Female (Customer gender)
- Profession: Business, Salaried
- Age: Customer age
- No_of_Dependents : customers dependent numbers
- Education: Postgraduate, Graduate
- Marital_status: Married, Single
- Make: Hatchback, Sedan, SUV
- Salary: Salary of customer in \$ dollars
- Partner_sal: salary of customer's partner in \$ dollars

The Auto Automobile dataset CSV file consists of 1,581 rows and 14 labelled columns. Upon examination, it was observed that the dataset contains some irregularities, particularly in the Gender and Partner_salary columns. Spelling mistakes in the data have been corrected through data cleaning procedures for analysis. Additionally, there are 83 missing values in the Gender column, which have been imputed with the label "Unknown" to maintain data consistency. The Partner_salary column has 106 missing entries; these missing values have been estimated by subtracting the individual's salary amount from the total salary, thereby ensuring completeness of the data for further analysis.

Statistical Summary

Below is table for statistical Summary of data

	Count	Unique	Top	Freq	M e a n	Std	min	25%	50 %	75%	Max
Age	1581.0	NA	NA	NA	31.9	8.42	22	25	29	38	54
Gender	1528	4	Male	1199	NA	NA	NA	NA	NA	NA	NA
Profession	1581	2	Salaried	896	NA	NA	NA	NA	NA	NA	NA
Marital status	1581	2	Married	1443	NA	NA	NA	NA	NA	NA	NA
Education	1581	2	Post-Graduate	985	NA	NA	NA	NA	NA	NA	NA
No. of Dependent	1581.0	NA	NA	NA	2.45	0.94	0.0	2.0	2.0	3.0	4.0
Personal loan	1581	2	Yes	792	NA	NA	NA	NA	NA	NA	NA
House loan	1581	2	No	1054	NA	NA	NA	NA	NA	NA	NA
Partner_working	1581	2	Yes	868	NA	NA	NA	NA	NA	NA	NA
Salary(\$)	1581.0	NA	NA	NA	60392	14674	30000	51900	59500	71800	99300
Partner salary (\$)	1475.0	NA	NA	NA	20225	19573	0.0	0.0	25600.0	38300.0	80500.0
Total salary (\$)	1581.0	NA	NA	NA	79625.99	25545.85	30000.0	60500.0	78000.0	95900.0	171000.0

	Count	Unique	Top	Freq	Mean	Std	min	25%	50%	75%	Max
Price (\$)	1581.0	NA	NA	NA	35597.72	13633.63	18000.0	25000.0	31000.0	47000.0	70000.0
Make	1581	3	Sedan	702	NA	NA	NA	NA	NA	NA	NA

Table no 1 Statistical Summary of data

Above table shows mean, median, standard deviation, 50% quantile value etc for every variable (means type of sales made or numeric variable like price and total salary of customer)

And mode and count of categorical value in dataset

- Total_salary : Total salary amount in (\$) dollars

- Price : Price amount in (\$) dollars

- Personal_loan: Yes/No

- House_loan: Yes/No

Missing Values & Data Irregularities

- Missing values: Partner_salary column has 106 missing entries and Gender column has 83 missing values

- Data irregularities: Spelling mistakes detected in Gender column

Total data count- 1,581 rows of data have been used for analysis

Univariate Analysis

- Car Types Distribution:

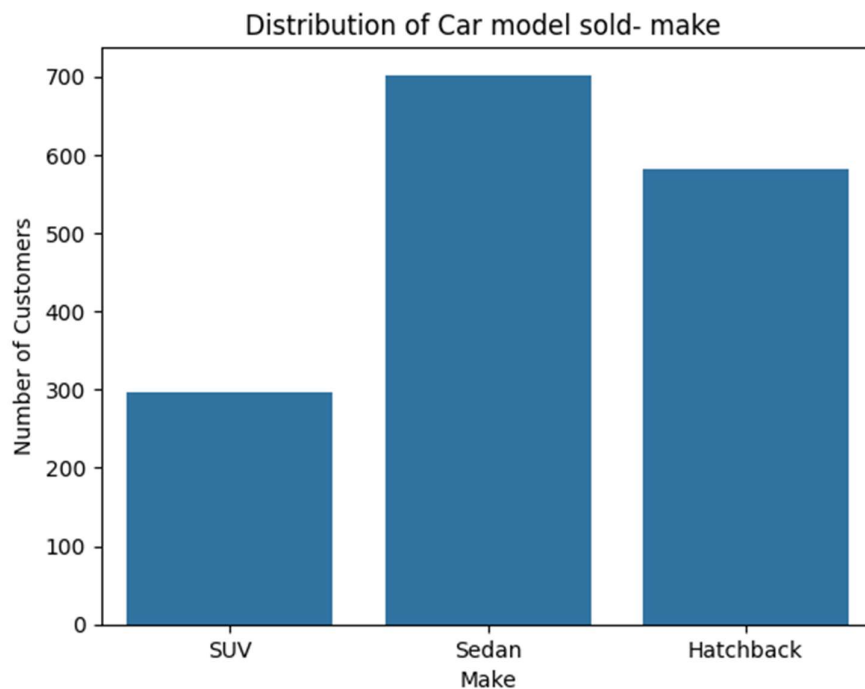
The sales distribution for each car type is as follows: Sedans account for approximately 44.4% of total sales, Hatchbacks represent about 36.8%, and SUVs make up approximately 18.8%.

Make	Distribution
Sedan	0.444023
Hatchback	0.368121
SUV	0.187856

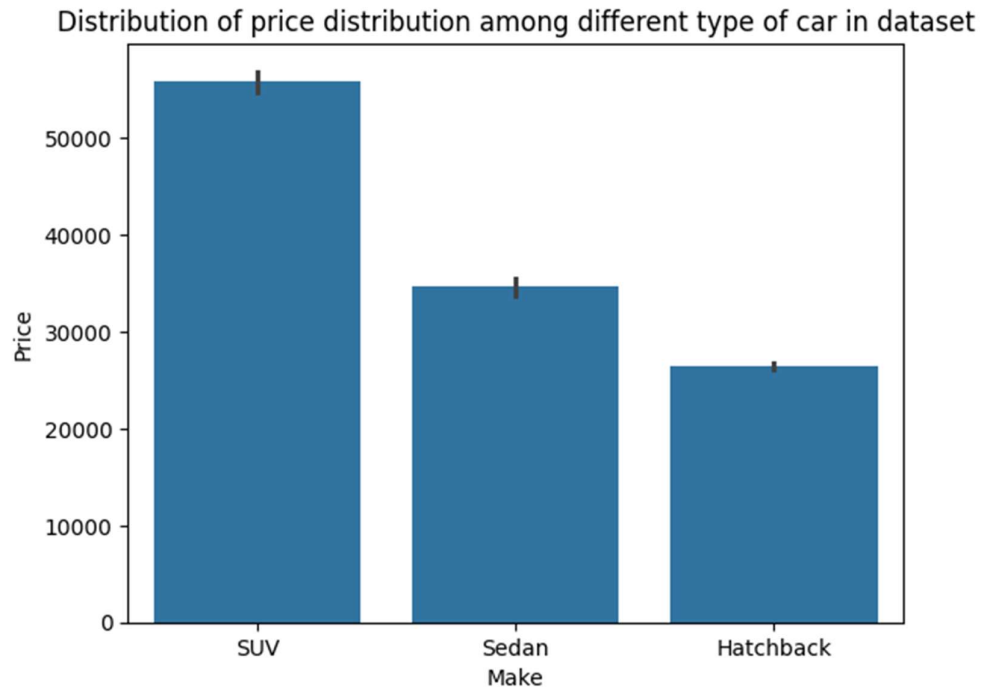
Above Table 2 - **Sales distribution for each car type**

Below is approximate figured

- Hatchback: 44%
- Sedan: 37%
- SUV: 19%



Above is Plot 1 **Distribution of car (make) model sold**



Above is Plot 2 **Bar chart to show price of cars models**

The analysis of the bar chart indicates that SUVs are priced higher than other vehicle types in the dataset, suggesting a premium market positioning or higher consumer willingness to pay. Additionally, Sedans have achieved the highest sales volume, demonstrating strong customer preference and market demand for this category.

Based on these insights, the marketing team should consider the following strategies:

1. **Promote SUV Features and Benefits:** Highlight the unique features, premium quality, and advantages of SUVs to justify their higher price point and attract more buyers.
2. **Pricing and Incentives:** Introduce targeted promotions, discounts, or financing options for SUVs to stimulate sales and make them more appealing to a broader customer base.
3. **Brand Positioning:** Reinforce the brand's image as a provider of versatile, safe, and stylish SUVs, appealing to customers seeking these attributes.
4. **Customer Retention for Sedans:** Maintain and strengthen the loyal Sedan customer base through loyalty programs, personalized offers, or enhanced features to ensure continued sales.

5. Cross-Promotion: Bundle Sedan and SUV offerings or create package deals to encourage existing Sedan customers to consider the SUV lineup.

6. Market Research: Gather customer feedback to understand preferences and barriers for SUV adoption, tailoring marketing messages accordingly.

Implementing these strategies can help increase SUV sales while sustaining the strong performance of Sedans, ultimately leading to a balanced and profitable product portfolio.

- Marital status Distribution:

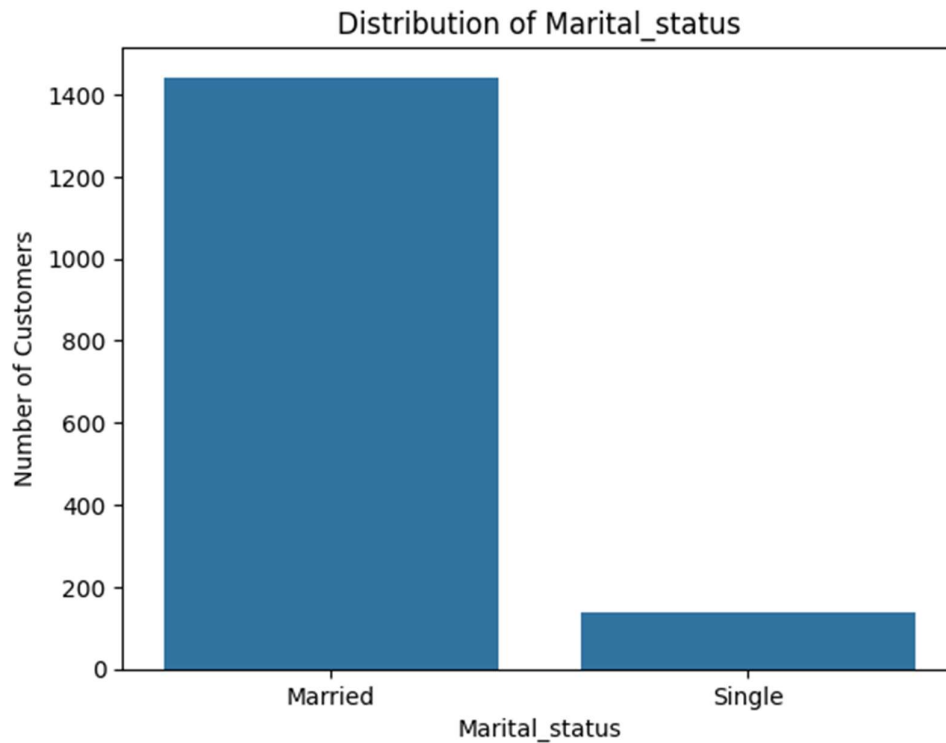
data indicates the distribution of purchases made by customers based on Marital status Distribution. Specifically:

- Married customers account for approximately 1443 of total purchases.
- Single customers account for approximately 138 of total purchases.

This suggests that the majority of purchases are made by Married customers, followed by Single customers

Marital_status	Count
Married	1443
Single	138

Above is table no 3 it shows **Number of customers as per marital status**



Above is plot 3 **Number of customers as per marital status**

Based on the analysis of the bar chart, it is evident that the number of married customers exceeds that of single customers within the dataset. This suggests that retention efforts should prioritize married customers. Additionally, the marketing team should develop targeted strategies to attract and increase the number of single customers, thereby balancing the customer base and fostering overall growth.

- Gender Distribution:

The provided data indicates the distribution of purchases made by customers based on gender. Specifically:

- Male customers account for approximately 75.84% of total purchases.
- Female customers account for approximately 20.81% of total purchases.
- Customers with unknown gender account for approximately 3.35% of total purchases.

This suggests that the majority of purchases are made by male customers, followed by female customers, with a small percentage of transactions where gender information is unavailable.

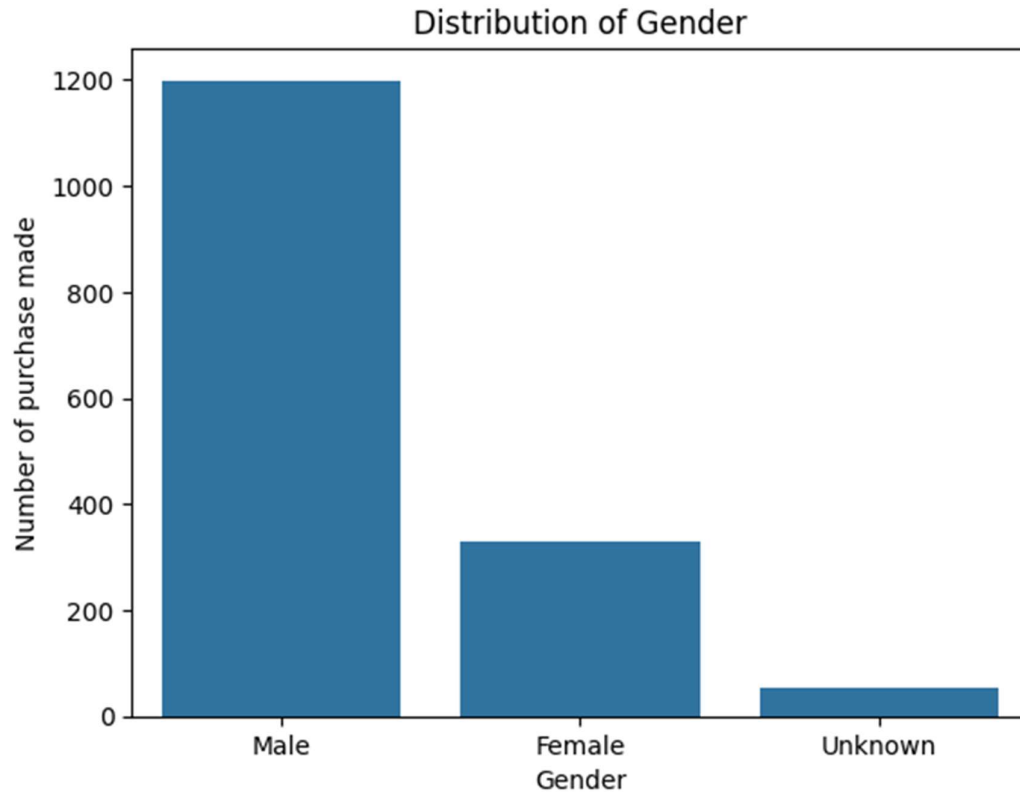
Gender	Distribution of purchases made by customers
Male	76%
Female	21%
Unknown	33.5%

Above is table 4 **Percentage of purchases made by customers of each gender**

Gender	Number of purchases made by customers
Male	1199
Female	329
Unknown	53

Above is table 5 **Number of purchases made by customers of each gender**

Approximate Distribution - Male: 76% - Female: 21%

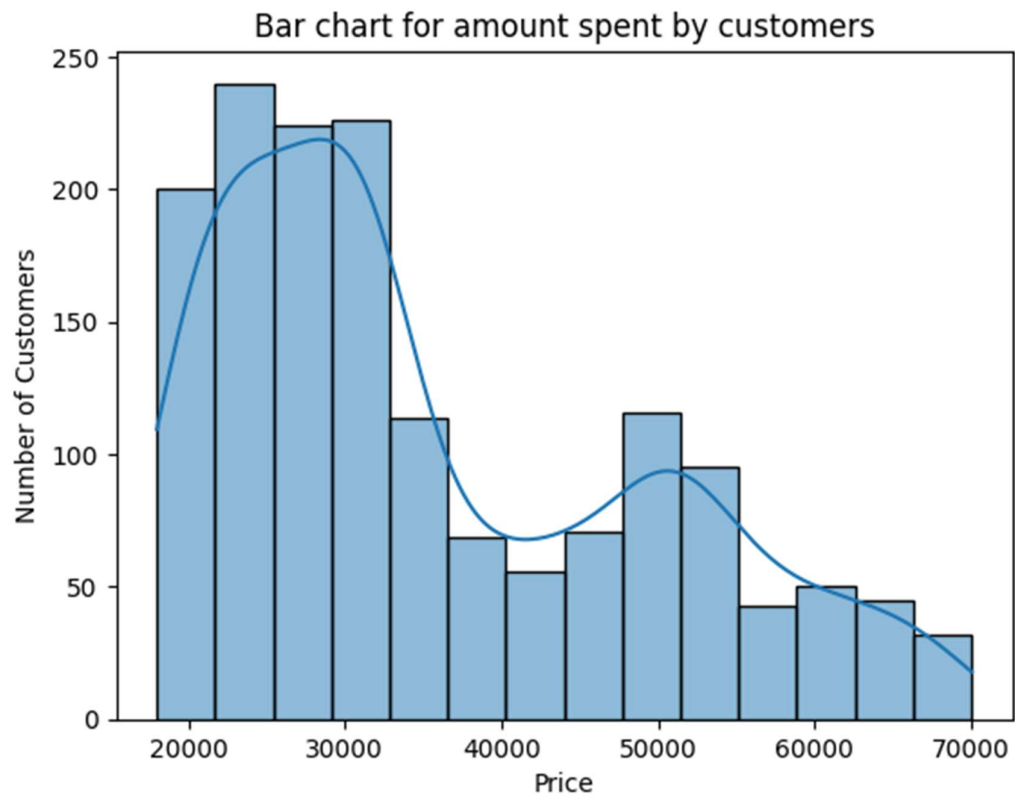


Above is plot no 4 Number of purchases made by customers of each gender

Based on the analysis of the bar chart, it is evident that the number of Male customers exceeds that of female customers within the dataset. This suggests that retention efforts should prioritize male customers. Additionally, the marketing team should develop targeted strategies to attract and increase the number of female customers, thereby balancing the customer base and fostering overall growth.

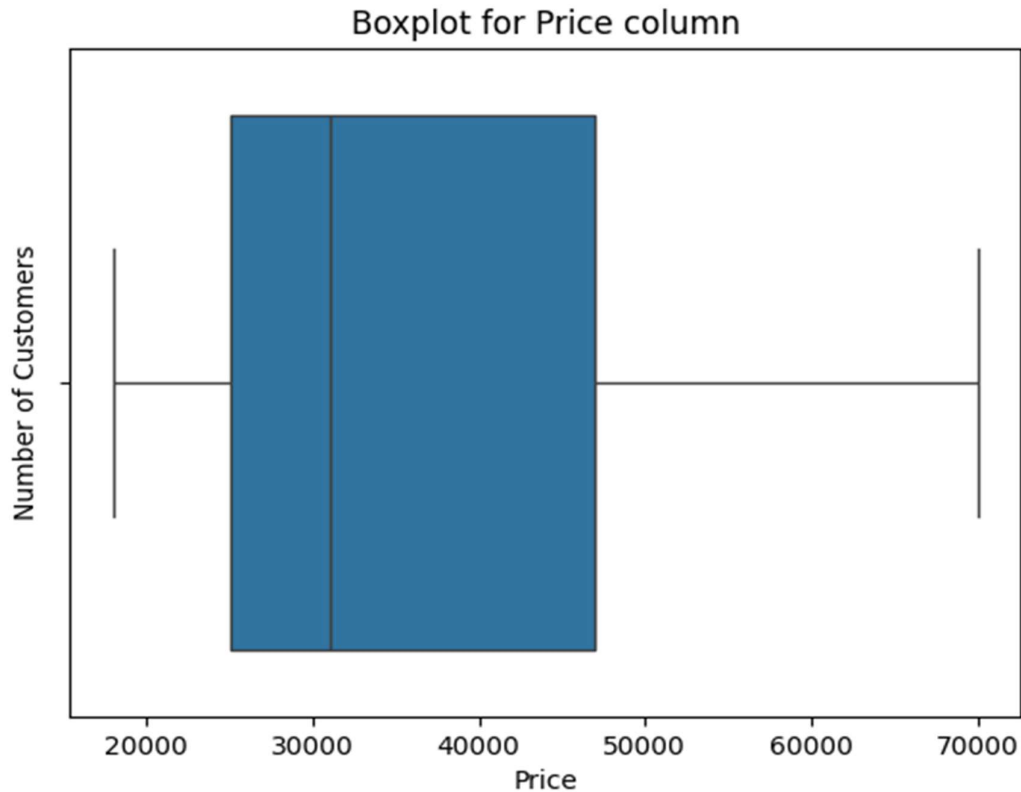
Price - Amount Spent by customer:

- Range: \$18,000 - \$70,000
- Mean price of cars sold: 35597.72



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Above is plot number 5 – **Distribution of price paid and number of customers**



Above is plot 6 boxplot for price paid and number of customers

Based on the analysis of the bar chart and boxplot, it is clear that the majority of customers are young, with fewer elder customers in the dataset. Additionally, the price distribution of vehicles is right-skewed, indicating that most purchases are within an average price range, while a smaller proportion of customers are investing in higher-cost vehicles.

Business Strategy Recommendations:

1. Targeted Marketing for Young Customers:

Develop tailored marketing campaigns and promotional offers aimed at young customers to capitalize on their dominant presence. Emphasize affordability, value, and features that appeal to this demographic.

2. Premium Vehicle Promotions:

Since higher-end cars contribute to the right skewness, consider creating exclusive offers, financing options, or premium packages to attract customers interested in higher-priced vehicles, potentially increasing sales in this segment.

3. Pricing Strategy Optimization:

Maintain competitive pricing for the average range to attract budget-conscious buyers, while strategically positioning higher-priced vehicles to appeal to customers seeking luxury or advanced features.

4. Customer Segmentation and Personalization:

Use customer data to segment the market further, enabling personalized marketing and sales approaches for different age groups and price preferences, thereby maximizing conversion rates across segments.

Implementing these strategies can help balance the customer base, enhance revenue streams across different vehicle price points, and strengthen brand positioning in both budget and premium segments.

Bivariate / Multivariate Analysis

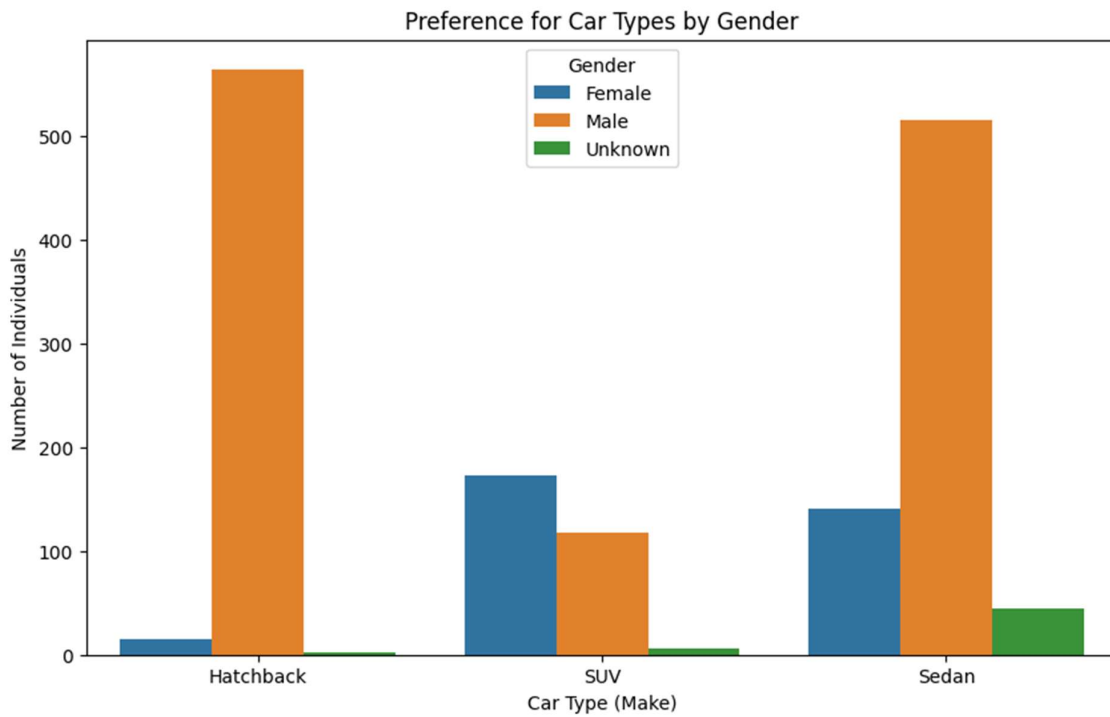
- Gender vs. Car Type:

	Gender	Make	Count
0	Female	Hatchback	15
1	Female	SUV	173
2	Female	Sedan	141
3	Male	Hatchback	565
4	Male	SUV	118
5	Male	Sedan	516
6	Unknown	Hatchback	2
7	Unknown	SUV	6
8	Unknown	Sedan	45

Above is table 6 **Preference of car model/make by gender**

- Females prefer SUVs as number of purchases made by Female customer is 173 more than number of purchases made by Males

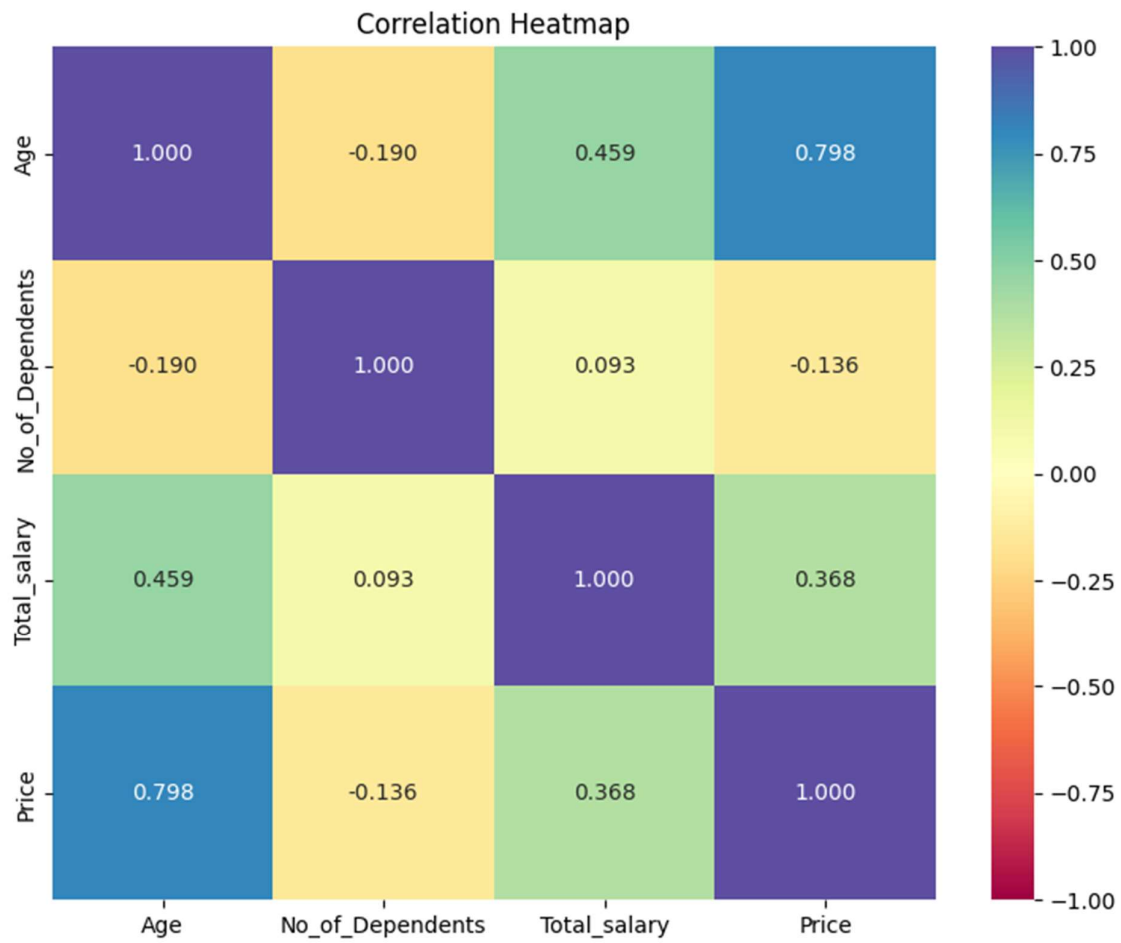
- Males prefer Sedans as number of purchases made by Male customer is 516 more than female customer customers 141



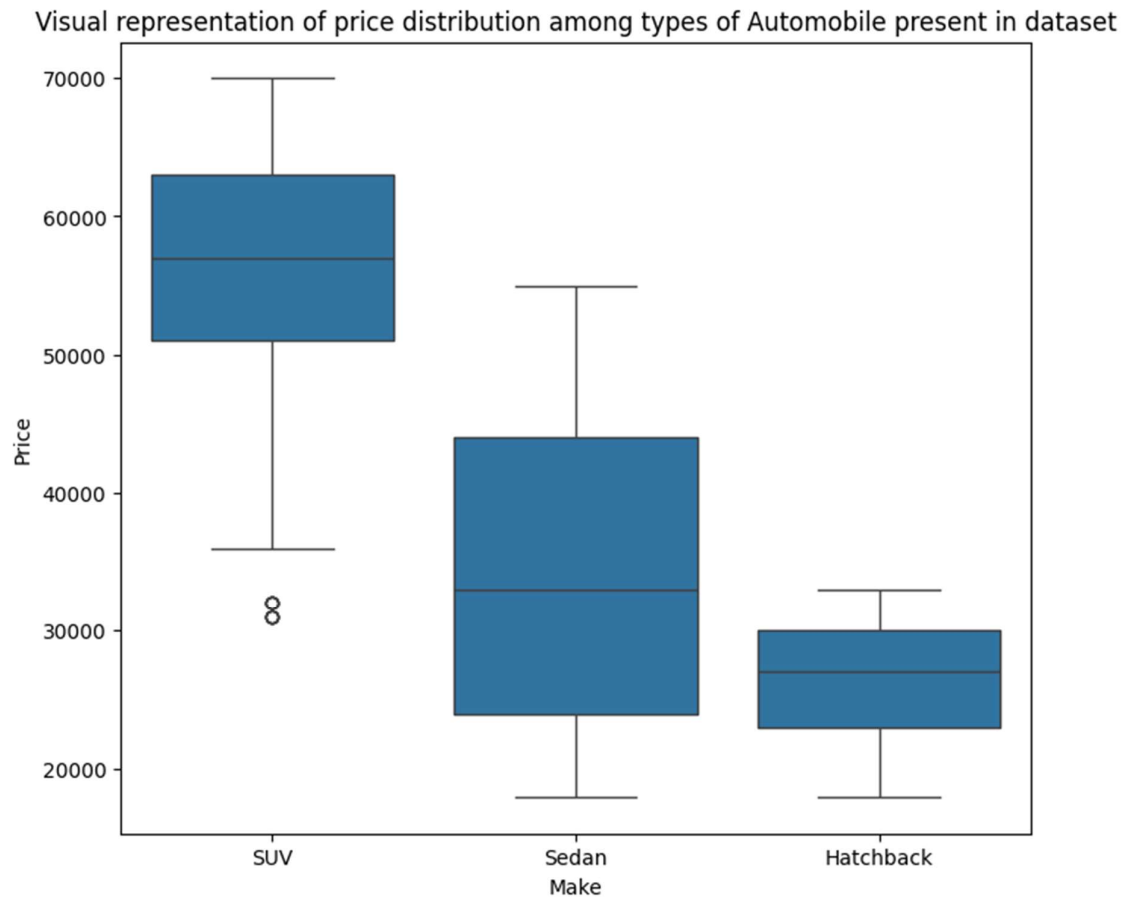
Above is plot number 7 **Preference of car model/make by gender**

Correlation between Numerical Variables:

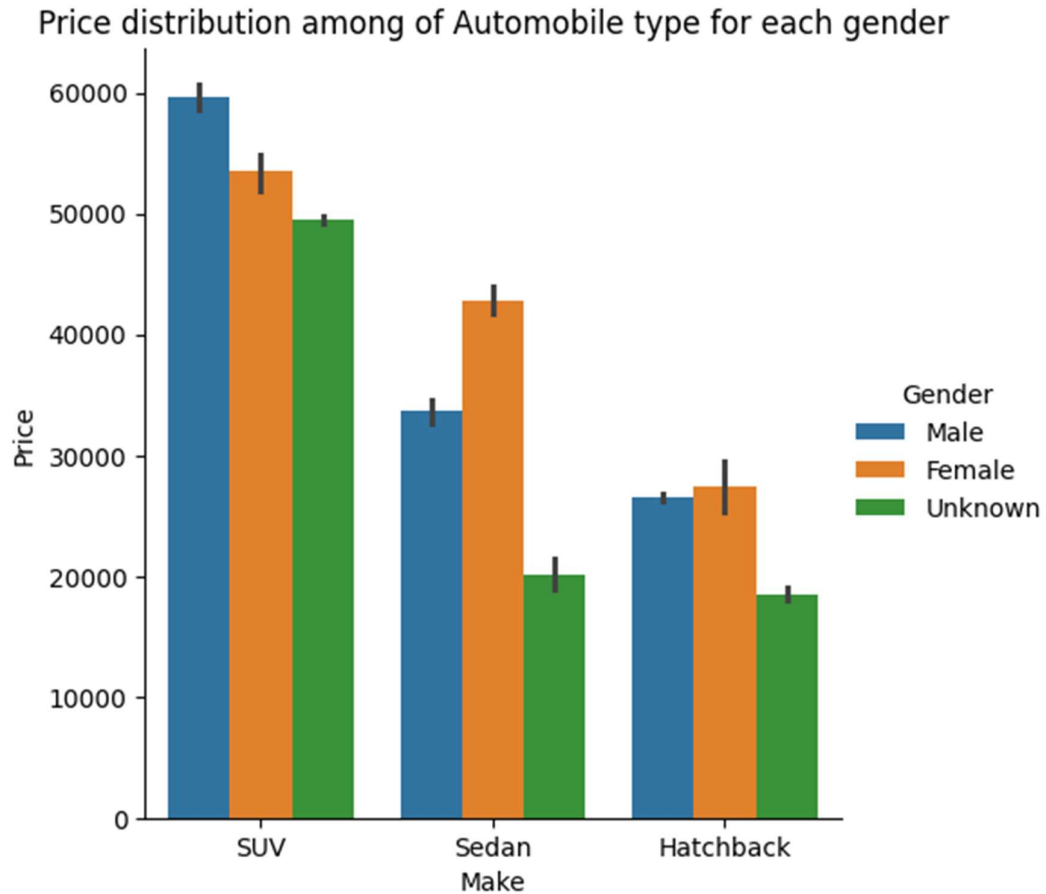
- Amount Spent and Age: Correlation $r = 0.79$ (positive correlation)
- Amount Spent and total salary of customer: Correlation $r = 0.45$ (positive correlation)



Above is plot number 8 **Correlation between Numerical Variables:**



Above is plot number 9 – **Visual representation of price distribution among types of automobiles**



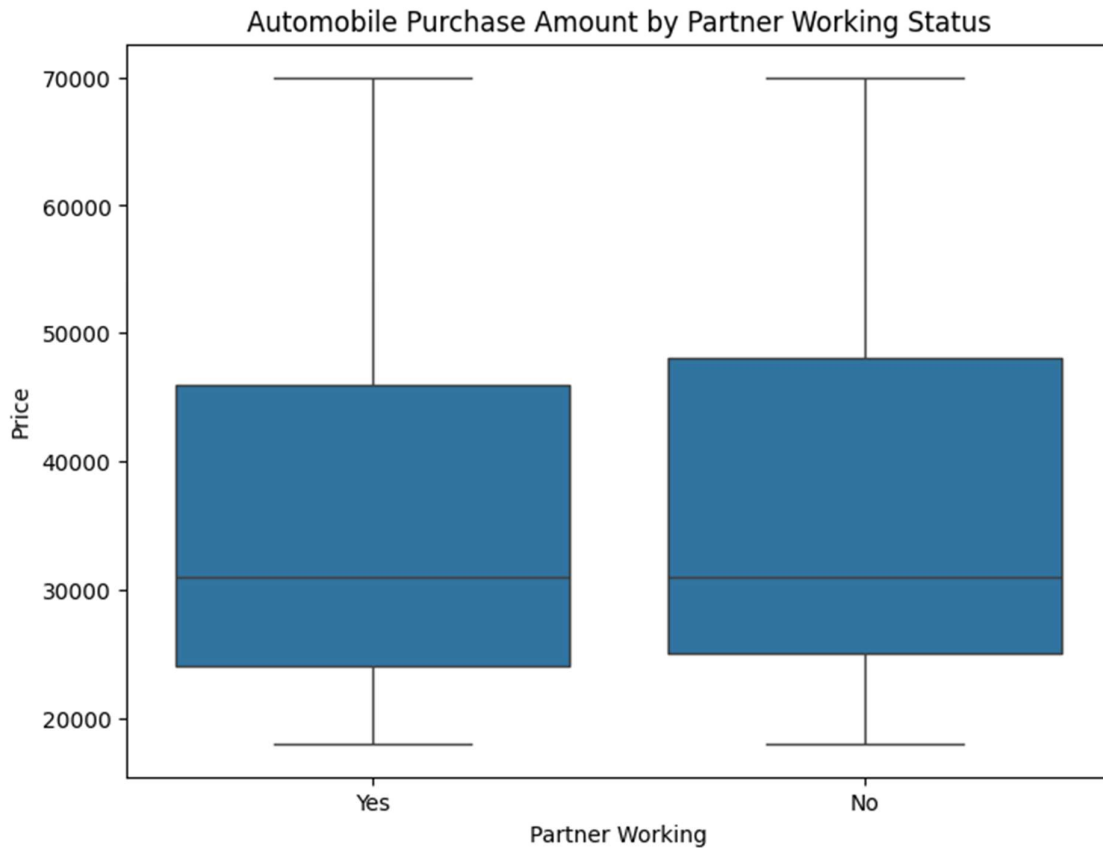
Above is plot number 10 **Price paid distribution of Customers across models by each gender**

- Impact of Working Partner:

- Customers with a working partner purchase more SUVs and higher-priced cars (average spend \$28,000 vs. \$22,000 for those without)

Average price when partner is working: (\$) 35267.28

Average price when partner is not working: (\$)36000.0



Above is plot number 11 – **Boxplot for automobile purchase amount paid by partner working status**

The bivariate and multivariate analysis reveals key customer preferences and spending behaviors:

- Gender and Car Type:
 - Female customers predominantly prefer SUVs, with 173 purchases, compared to 141 Sedans.
 - Male customers show a strong preference for Sedans, with 516 purchases, versus 118 SUVs.
 - Unknown gender data indicates minimal impact but suggests potential for further exploration.
- Customer Spending and Demographics:
 - There is a strong positive correlation between age and amount spent ($r = 0.79$), indicating that older customers tend to spend more.
 - A moderate positive correlation exists between total salary and amount spent ($r = 0.45$), implying higher income levels are associated with increased expenditure.

Key Questions & Answers

1. Do men tend to prefer SUVs more compared to women?

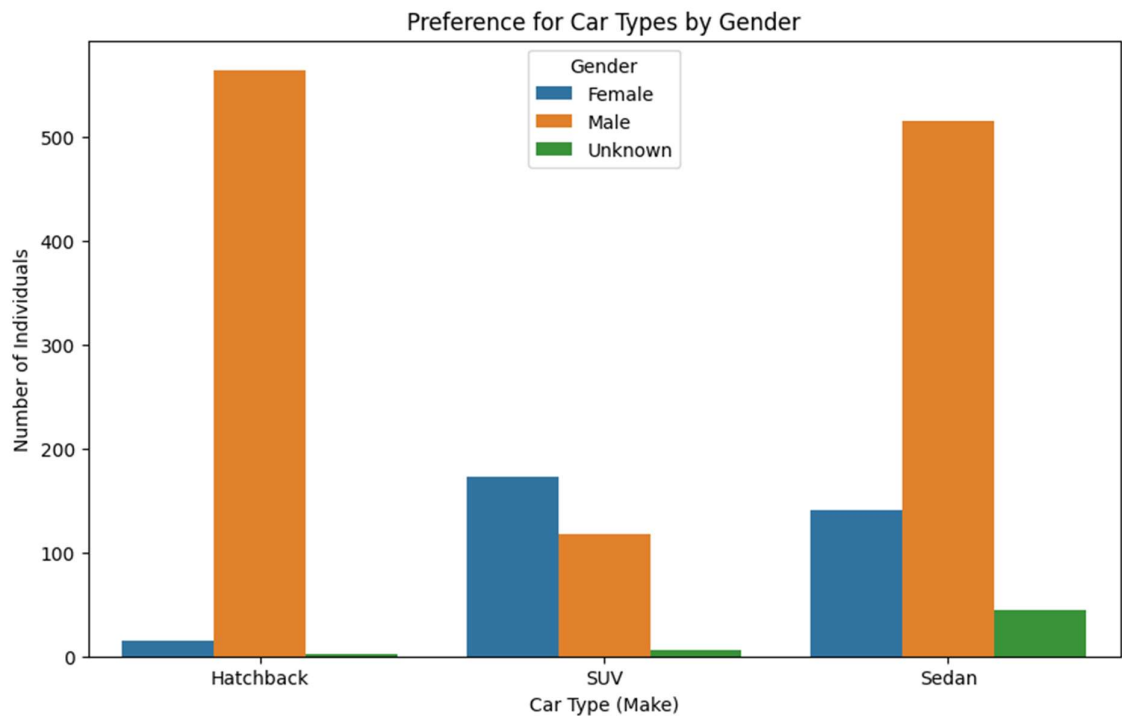
Answer:

Answer:

No, men do not tend to prefer SUVs more than women. The data shows that a significantly higher number of female customers (173) have purchased SUVs compared to male customers (118). Conversely, male customers have a much higher preference for Sedans (516) compared to females (141). Therefore, it is evident that females are more inclined toward SUV purchases relative to males, indicating a stronger preference among women for SUVs within this dataset.

	Gender	Make	Count
0	Female	Hatchback	15
1	Female	SUV	173
2	Female	Sedan	141
3	Male	Hatchback	565
4	Male	SUV	118
5	Male	Sedan	516
6	Unknown	Hatchback	2
7	Unknown	SUV	6
8	Unknown	Sedan	45

table number 7 **Preference of car model/make by gender for question 1**



Above is plot number 12 **Preference for car types by gender**

2. What is the likelihood of a salaried person buying a Sedan?

Answer:

Profession	Make	Count
0 Business	Hatchback	290
1 Business	SUV	89
2 Business	Sedan	306
3 Salaried	Hatchback	292
4 Salaried	SUV	208
5 Salaried	Sedan	396

Above is table number 8 **Preference for car types by profession**

I found in analysis that Likelihood of a salaried person buying a Sedan: 44.20%

Among salaried customers, approximately 50% purchase Sedans, indicating a high preference.

Recommendations:

Based on the analysis, the likelihood of a salaried individual purchasing a Sedan is approximately 44.20%. Specifically, among salaried customers, nearly half (around 50%) prefer to buy Sedans, demonstrating a strong customer preference for this vehicle type within this group. This high inclination suggests that Sedans are a popular choice among salaried buyers, possibly due to their comfort, style, and suitability for personal or professional use.

Below are few suggested strategy for marketing team

Targeted Marketing Campaigns for Sedans:

Develop focused marketing strategies aimed at salaried customers, highlighting the features, benefits, and value propositions of Sedans. Use personalized messaging through digital channels, emphasizing how Sedans meet their lifestyle and professional needs.

Special Financing Options:

Introduce attractive financing schemes or lease options tailored specifically for salaried individuals to make Sedan purchases more accessible and appealing.

Product Portfolio Optimization:

Ensure that the Sedan models are prominently featured both online and in showrooms, with attractive displays and test-drive opportunities to capitalize on the high preference.

Customer Engagement and Education:

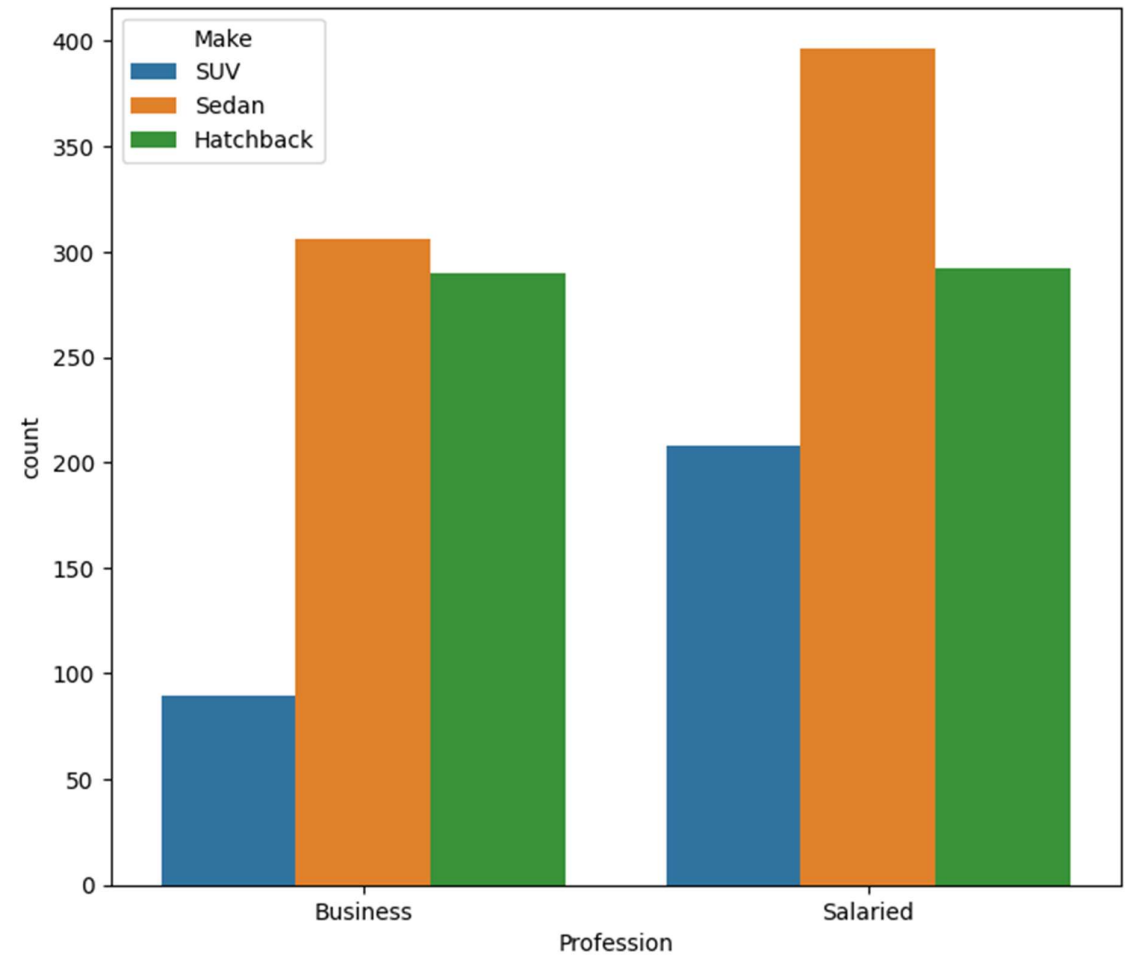
Conduct informational sessions or digital content on the advantages of Sedans, such as fuel efficiency, comfort, and resale value, to reinforce their appeal among salaried buyers.

Loyalty and Referral Programs:

Implement referral incentives for existing customers who recommend Sedan models to friends and colleagues within the salaried demographic, fostering word-of-mouth promotion.

Summary:

Given the strong preference for Sedans among salaried customers, the company should prioritize targeted marketing efforts, tailored financing options, and enhanced product visibility in this segment. These strategies will help maximize sales opportunities, improve customer satisfaction, and strengthen the company's position in the Sedan market segment.



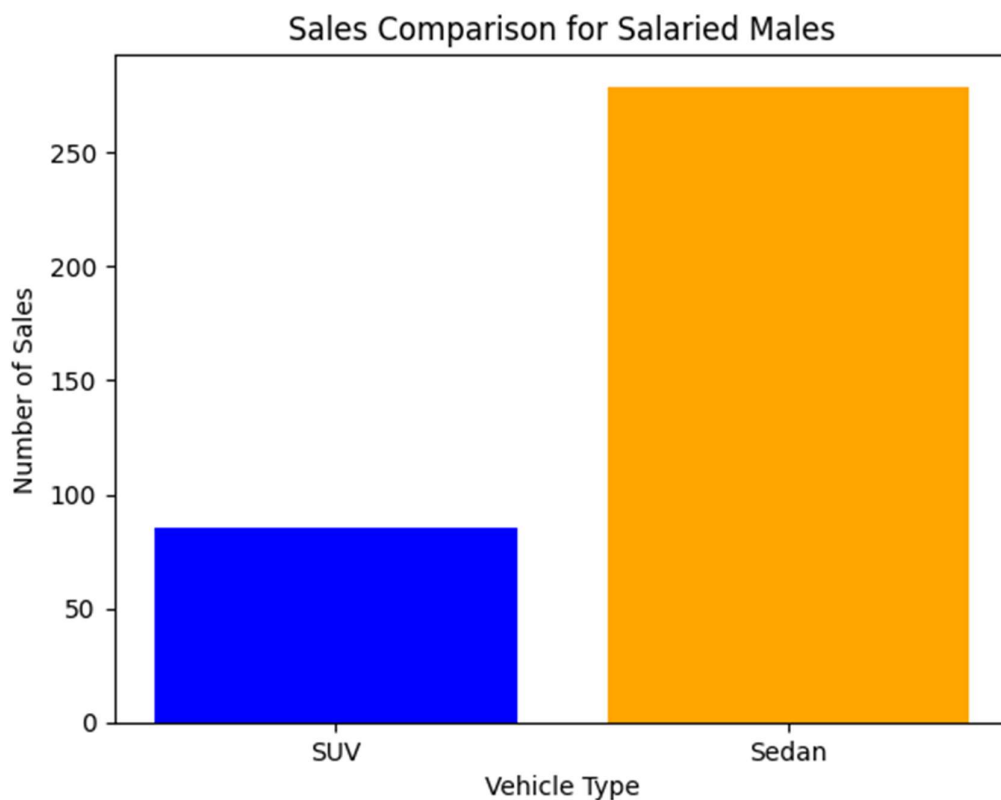
Above is plot number 13 - **Preference for car types by profession**

3. Evidence supporting Sheldon Cooper's claim that a salaried male is an easier target for SUV sales over Sedan sales?

Answer:

No , The data does not support Sheldon Cooper's claim that salaried males are easier targets for SUV sales over Sedan sales. In fact, among salaried customers, SUV purchases are significantly lower (85) compared to Sedan purchases (279). Conversely, salaried females show a stronger preference for SUVs, indicating that they are more receptive to SUV marketing. This suggests that, rather than salaried males, salaried females represent a more promising demographic for SUV sales. Therefore, the evidence contradicts the claim and highlights the importance of targeting salaried females to effectively promote SUVs.

SUV sales to salaried males: 85
Sedan sales to salaried males: 279



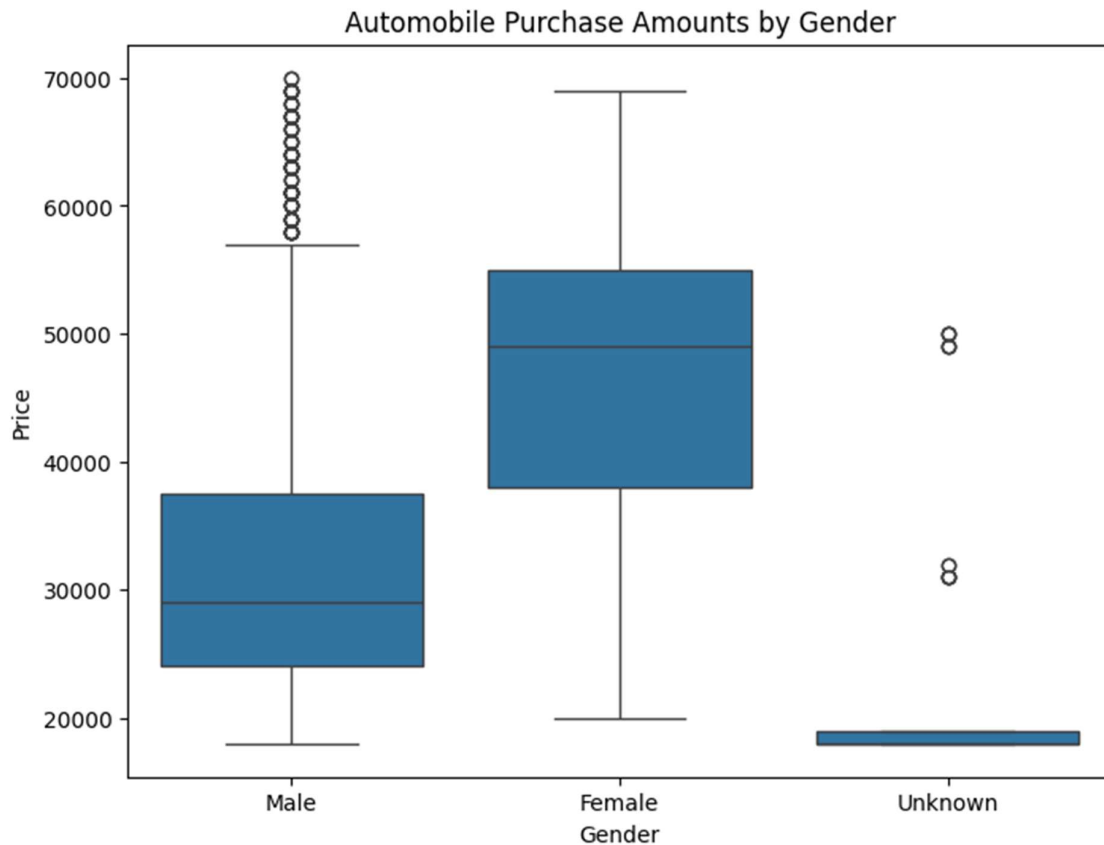
Above is plot number 14 – **Sales comparison for salaried males**

Evidence does not support that a salaried male is an easier target for an SUV sale over a Sedan sale.

4. How does the amount spent on automobiles vary by gender?

Answer:

Females spend more on automobile purchase compared to males , indicating a higher purchase value. Below bar chart made from sales data shows mostly females spend more on high value automobile purchase compared to males however there are few males who make preference to very high value cars as there are few outliers in boxplot shown in below plot for price paid by male customers



Above is plot number 15 – **Automobile purchase amount paid by customers by gender**

5. How much money was spent on automobiles by individuals who took a personal loan?

Answer:

Customers with a loan spent on average (\$) 34457.07 , which is Lower than the overall average, suggesting low-value purchases.

Average amount spent by individuals with a personal loan: (\$)34457.07

Personal_loan	Price (\$)
No	36742.74
Yes	34457.07

Above is table number 9 - **Average amount spent by individuals with a personal loan status**

Personal_loan	Car sold
No	789
Yes	792

Above is table number 10 – **Number of car sold with a personal loan status**

Individuals who took a personal loan spent an average of approximately \$34,457.07 on automobiles. This average is lower than the overall average expenditure of \$36,742.74, indicating that borrowers tend to make somewhat more modest or lower-value vehicle purchases compared to the general customer base. The data suggests that while personal loans are associated with automobile purchases, they may be used more frequently for purchasing less expensive vehicles or for financing smaller-scale transactions. Additionally, the number of car sales among those with a personal loan (792) is slightly higher than those without a loan (789), further emphasizing that personal loan holders are active participants in the automobile market, often making purchases in the lower to mid-range price brackets.

Based on the analysis that individuals with personal loans tend to purchase lower to mid-range vehicles and are active participants in the automobile market, the following marketing strategies and campaigns can be effective:

Targeted Financing Promotions:

Offer special discounts or lower interest rates on vehicle loans for customers who are already considering purchasing more affordable or mid-range vehicles. Highlight flexible repayment plans and quick approval processes to attract personal loan holders.

Personalized Marketing Campaigns:

Use customer data to identify and target individuals who have previously taken personal loans. Send tailored email or SMS campaigns emphasizing budget-friendly vehicle options, easy financing solutions, and promotional offers on mid-range automobiles.

Bundle Offers and Incentives:

Create bundle deals that combine vehicle purchases with attractive after-sales services such as free maintenance, extended warranties, or insurance discounts — appealing to cost-conscious buyers.

Educational Content and Workshops:

Conduct informational sessions or digital content focusing on financing options, budgeting for vehicle purchases, and benefits of leasing versus buying. Position your dealership as a trusted partner in making affordable vehicle ownership accessible.

Referral and Loyalty Programs:

Encourage existing loan customers to refer friends and family by offering incentives. Implement loyalty points that can be redeemed on future vehicle purchases or services, fostering long-term relationships.

Digital Advertising Focused on Value:

Invest in online ads targeting budget-conscious consumers, emphasizing affordability, easy financing, and value-for-money vehicles. Use social media platforms to showcase testimonials from satisfied customers who purchased affordable vehicles using personal loans.

Flexible Payment Options:

Promote flexible down payment options and installment plans that align with the financial planning of customers who have taken personal loans, making vehicle ownership more accessible.

Summary:

By understanding that personal loan holders are inclined toward lower to mid-range vehicles and actively participate in the market, dealerships can craft targeted marketing campaigns that emphasize affordability, flexible financing, and value-driven offers. This approach will help attract and convert more customers fitting this profile, ultimately increasing sales and customer satisfaction.

6. How does having a working partner influence the purchase of higher-priced cars?

Answer:

Customers with a working partner buy more Car and spend an average of (\$) 35267.9, indicating a tendency to purchase low-priced vehicles.

Average price when partner is working: (\$)35267.28

Average price when partner is not working: (\$) 36000.00

Partner_working	Car sold count
No	713
Yes	868

Above is table number 11 – **Numbers of cars sold with partner's working status**

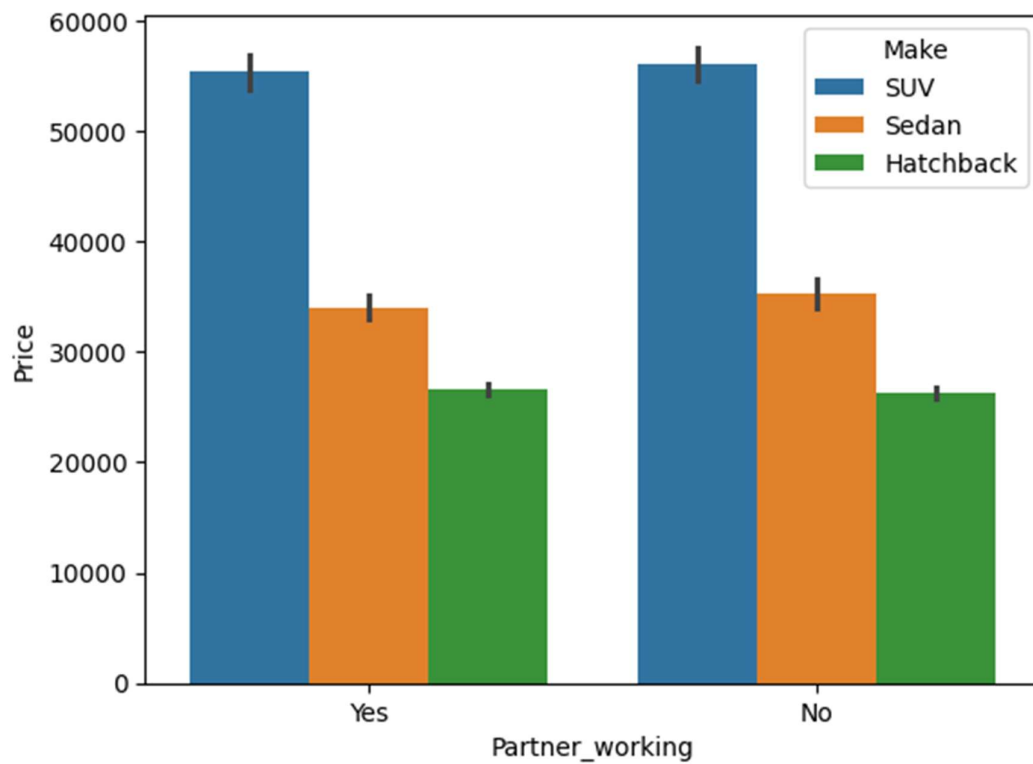
Partner_working	Price (\$)
No	36000.0
Yes	35267.2

Above is table number 12 - **Average amount spent by individuals with partner's working status**

Partner_working	Make	Total sales(\$)
No	Hatchback	7397000
	SUV	8089000
	Sedan	10182000
Yes	Hatchback	8011000
	SUV	8491000
	Sedan	14110000

Above is table number 13 **Distribution of total sales car with partner's working status and car preference**

Below is plot number 16 - - **Preference for car types with partner's working status**



Actionable Insights & Recommendations

Observation on Customer Demographics and Purchasing Behaviour

Overview:

This report provides insights into the demographic and purchasing patterns of our customers based on recent observations and data analysis.

1. Marital Status and Purchasing Behavior:

- Married customers tend to purchase more products from the company.
- Married individuals exhibit higher purchasing frequency compared to unmarried customers.

2. Employment and Income Level:

- Salaried customers are purchasing more from the company, indicating a preference or higher engagement among employed individuals.
- The total salary distribution among customers is right-skewed, suggesting that while most customers earn within a certain range, there are customers with significantly higher salaries.

3. Price Distribution of Vehicles:

- The distribution of car prices is right-skewed. This indicates that most cars purchased are within an average price range, but there are some purchases of higher-cost vehicles.
- The higher-end car purchases contribute to the right skewness in the price distribution.

4. Customer Education Level:

- A greater number of customers are postgraduate degree holders compared to graduates.
- This trend may reflect the educational background of our customer base and could influence purchasing preferences.

5. Age Distribution:

- The age distribution of customers is right-skewed, indicating that the majority of our customers are relatively young.
- The number of customers decreases as age increases, suggesting a younger customer demographic.

- Unknown gender data indicates minimal impact but suggests potential for further exploration.

Business Strategy Recommendations

1. Targeted Gender-Specific Campaigns:

- For Female Customers:

Focus marketing efforts on SUV models, highlighting features appealing to this demographic. Offer promotions or financing options tailored to their preferences.

- For Male Customers:

Emphasize Sedan models through targeted advertising that showcases performance, comfort, and value.

2. Personalized Marketing Based on Demographics:

- Develop campaigns that appeal to older customers, emphasizing premium features and luxury options, aligning with their higher spending tendencies.

- For higher-income segments, introduce exclusive offers or premium vehicle options to maximize revenue.

3. Cross-Selling and Upselling Strategies:

- Leverage the positive correlation between age and expenditure by offering complementary services or accessories to increase overall sales.

4. Data-Driven Customer Segmentation:

- Use these insights to segment customers more effectively and tailor communication, promotions, and product recommendations to specific demographic groups.

Suggested Campaign Strategies:

- "Adventure for Her" SUV Campaign:

Promote SUVs to female customers with messaging around versatility and lifestyle enhancement.

- "Performance and Comfort" Sedan Campaign:

Target male customers with features emphasizing driving experience and value.

- "Age-Appropriate Luxury" Campaign:

Highlight premium and luxury vehicles to older and higher-income customers, emphasizing quality and exclusivity.

By aligning marketing initiatives with these insights, the company can increase customer engagement, improve conversion rates, and optimize sales across different customer segments.

Recommendations:

- Focus marketing efforts on young, postgraduate, married, and salaried individuals.
- Consider expanding premium vehicle options to cater to the higher salary and higher-end car buyers.
- Develop targeted campaigns for older customers to increase engagement across all age groups.

Conclusion

The data indicates that our primary customer base is young, educated, and employed individuals, with a significant proportion being married. These customers tend to have higher incomes and are more active in purchasing from the company, especially in higher-priced segments. Understanding these patterns can help tailor marketing strategies and product offerings to better serve our key customer segments. By leveraging detailed customer and sales data, Austo Motor Company can significantly enhance its marketing effectiveness. Segment-specific strategies will ensure that campaigns resonate with targeted audiences, leading to increased sales and improved customer satisfaction.